What makes a good research question?

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Overview

A good research question:

- Why do we need one?
- What does it look like?
- How do you know you got one?

What about a hypothesis?

The power of the research question

- It defines an investigation
- It sets boundaries
- It provides direction
- It acts as a frame of reference

Topic

Each group project will be on the following topic: *Is mind wandering related to screen time?*
The best research topics are:

- of interest to you
- significant to the field
- can provide answers that can be acted upon
- in alignment with the assignment

Topic: Ask yourself

1. Is this topic of interest to me?
2. Is this topic of significance?
3. Will research into this topic be useful?
4. Does research into this topic meet the course requirements?

Articulating your research question: clarity and specificity

Clear

- Uses plain, unambiguous language
- Defines key terms (what exactly do you mean by “mind wandering” or “screen time”?)
- Can be understood by someone outside your topic
- Asks one question, not several

Specific

- Names the population (who?)
- Names the variables or concepts (what?)
- Sets the context or scope (where? when?)

- Is narrow enough to answer within your word count and time

From vague to focused

✗ *Does social media affect mental health?*
Which platforms? Which aspects of mental health? Whose?
✗ *How does social media affect young people's mental health and academic performance and sleep?*
Too many questions at once
✓ *Is daily short-form video viewing (e.g. TikTok, Instagram Reels) associated with symptoms of anxiety in first-year undergraduate students?*

- Behaviour: short-form video viewing, defined as watching clips under 3 minutes on a scrolling feed
- Population: first-year undergraduates
- Outcome: symptoms of anxiety
- Clear population, concepts and scope; can be answered with a defined study

Check your own question

Is it yours?

- Am I curious about the answer?
- Could I stay interested for the whole project?
- Can I manage any biases/subjectivities I may have?

Does it fit the field?

- Does it connect to existing research?
- Would other researchers see the findings as relevant?

Are the terms well defined?

- Could I explain each key term in one sentence?

- Would someone else interpret them the same way?

Any unchecked assumptions?

- Am I assuming a link or direction (e.g. that A causes B)?
- Am I assuming my population or measure is straightforward?
- What would I have to take for granted for this question to make sense?

Is it doable?

- Can I access the participants, data or sources I need?
- Can I answer it within my time, word count and resources?
- Could I describe what evidence would count as an answer?

What about a hypothesis?

A hypothesis is a logical conjecture about the nature of relationships between two or more variables expressed in the form of a testable statement.
You need:

1. a clearly defined research question
2. variables to explore and
3. a hunch about the relationship between those variables that
4. can be tested

Can you write a hypothesis for the following question?

1. Do pre-school children who eat fish a minimum of three times a week perform better in dexterity tests than non-fish eaters?

Can you write a hypothesis for the following question?

2. How do returned service people with post traumatic stress disorder (PTSD) deal with questions about their condition?

How to write your hypotheses

Hypotheses are statements about the **population**, not just your sample. You need a null (H0) and an alternative (HA).

	Relationship (correlation)	**Group difference**
H0	There is no relationship between X and Y	There is no difference between Group A and Group B on Y
HA, non-directional	There is a relationship between X and Y	There is a difference between Group A and Group B on Y
HA, directional (positive / higher)	There is a positive relationship between X and Y	Group A will score higher than Group B on Y
HA, directional (negative / lower)	There is a negative relationship between X and Y	Group A will score lower than Group B on Y

- Replace X, Y and the group names with your **actual variables**
- Choose directional or non-directional **before** collecting data, and justify directional predictions with theory or previous findings

Activity: Discuss, agree, swap and check

1. **Go and sit in your groups.** Take a handout. You need 1 handout for every 2 or 3 people in your group. Write your group's name or number in the "Author" box.
2. **Agree on a question (~10 min).** Discuss ideas and agree on one draft research question for your group. Write it in Section 1, with one line on who or what it's about. Everyone should be able to explain why you chose it.
3. **Pair up across groups (~1 min).** Two (or three) of you from each group will go and join two (or three) people from another group.
4. **Swap (~1 min).** In your new foursome, swap handouts with the other pair, so each pair holds the other group's question.
5. **Review (~5 min).** With your partner, read the other group's question and complete Section 2 on their handout:
    - Tick **Yes** or **Unsure** for each check, and add a note where you can.
    - Write **one assumption** you can spot in the question.
    - Write **one thing you'd need to know** to answer it.

6. **Hand back and talk (~4 min).** Return the handout to the group that wrote it. Talk it through in your foursome:
    - What surprised each pair?
    - How did the reviewers interpret the key terms?
7. **Return and feed back (~7 min).** Go back to your original group. Share the feedback with the rest of the group. Use the notes on your handout.
8. **Revise together (~8 min).** As a group, rewrite your question with **at least one change** for clarity or specificity. Complete Section 3 to record what you changed and why.